

PARTICULAR SPECIFICATION P11

Initial Operation (IO) Periods

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## P11.1 OVERVIEW

- (a) As the TWRP liquids modules will be set to work progressively, the biosolids building, and digesters will also need to be brought into service progressively and, during such periods, the Contractor shall operate Digestion Pre-treatment and Siderstream Nitrogen Removal Facilities together with biosolids contractor to facilitate this approach. These are designated the Initial Operation (IO) periods under Part 1.
- (b) The IO period shall commence when either or both facilities (Digestion Pre-treatment facility or Siderstream Nitrogen Removal facility) successfully completed Stage 3a Process Commissioning based on the requirements stipulated in General Specification G2.26.9(q) and S.O.'s approval.
- (c) The IO periods include these key activities:
  - (i) Adjustment to new loads and flows; and
  - (ii) Plant performance testing.
- (d) All plant and equipment may be subject to initial operation by the Contractor until the whole of the Works has successfully completed the Plant Performance Test Period (PPTP). This is referred to as IO periods and is described in Table P11-1.
- (e) Table P11-1 describes the IO periods expected for the facilities. The Contractor shall note that the nominal duration is indicative as it is dependent on the Start and End of IO period.
- (f) Once the PPTP for the whole of the Works has been successfully completed, the S.O. will issue the Certificate of Substantial Completion (CSC) for the whole of the Works and this will initiate the start of the Operations and Maintenance period described in Particular Specification P12 Operations and Maintenance.
- (g) The Greasy Waste Receiving Facility will be available soon after the commissioning of THP. When GWRF is commissioned, the Contractor shall coordinate with C4A contractor to commission and integrate the HW supply to GWRF and ensure the systems are able to achieve the design objectives.
- (h) Before Digester Pre-treatment system is completed, the WASD will be diverted to PSI centrifuge temporarily by C4A contractor. Once THP is available for commissioning, the Contractor shall coordinate closely with the C4A contractor to commission the pre-dewatering centrifuge, biosolids digestion and digester pre-treatment as one integrated system.

**Table P11-1: C4B Initial Operation Period**

| IO Period | Start of IO Period | End of IO Period                                                          | Nominal Duration (months) | Facilities subject to Initial Operation                                        |
|-----------|--------------------|---------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------|
| IO1       | As per P11.1(b)    | Either Start of IO2 or Until substantial completion of the Whole of Works | 5                         | Either Digestion Pre-treatment facility or Sidestream Nitrogen Removal         |
| IO2       | As per P11.1(b)    | Until substantial completion of the Whole of Works                        | 2                         | Both Digestion Pre-treatment facility and Sidestream Nitrogen Removal facility |

- (i) During IO periods, the Contractor shall:
- (i) Ensure that all operational plant and equipment is properly maintained in accordance with manufacturers' recommendations. This shall include preventive and breakdown maintenance for the plant and equipment and shall including the provision of all special tools, instrumentation and analysers;
  - (ii) Ensure the support of the manufacturer and where necessary, engage the manufacturer's representatives to carry out specialist servicing and maintenance of equipment;
  - (iii) Electricity and water will be provided by Other contractors during IO periods; and
  - (iv) The Contractor shall include all other costs for operations and maintenance of the facilities during IO periods. This will include but not limited to the following operations and maintenance costs:
    - 1) Labour;
    - 2) Maintenance;
    - 3) Chemicals (If any);
    - 4) Grease, lubricating oil, diesel, instrumentation consumables, spare parts;
    - 5) Sampling and analysis for investigation purpose and as specified in Appendix P12A;
    - 6) Replenishment of seed sludge for Sidestream Nitrogen Removal in the event of loss of Anammox sludge in the Reactors due to operational failures (If Cross-seeding is not possible); and
    - 7) Certification for all statutory testing and requirements.

## P11.2 OVERVIEW OF SCOPE OF WORK DURING EACH IO PERIOD

- (a) This Specifications shall be read in conjunction with Particular Specification P12 Operations and Maintenance.

- (b) The Scope of Work to be carried out during the IO periods shall be as defined for the OM periods in Particular Specification P12 Operations and Maintenance.
- (c) The Contractor shall refer to Particular Specification P12 Operations and Maintenance for the personnel required during the IO periods.
- (d) The Contractor shall note that maintenance carried out during the IO Periods will eventually be continued into the operations and maintenance (OM) periods, described in Particular Specification P12 Operations and Maintenance, and that maintenance regimes during the IO periods must be planned accordingly. This will need to be agreed with the S.O.

**END OF SECTION**